Exp No: 04

SETTING UP A LOCAL AREA NETWORK (LAN)

Aim

To design, set up and configure a LAN connecting four computer through a network switch, assign static IP address to all devices, verify connectivity using ping command and demonstrate resource / data sharing across the world.

Equipment Required:

SNO	Equipment	Quantity	Specification
1	Personal computers (PC)	4	Ethernet enabled
2	Network switch	1	8/16/24 port, unmanaged / managed
3	Ethernet cables	4	Straight, RJ-45 connectors

Theory:

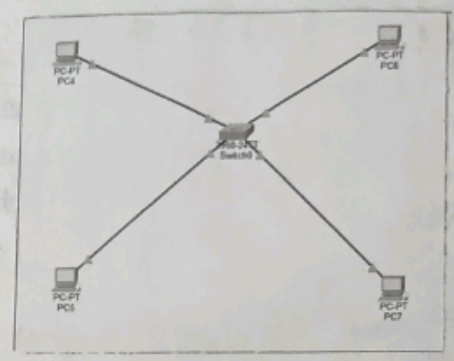
A local Area Network is a Network that connects computers and devices with a limited geographical area such as home, office or laboratory allowing them to share resources such as files, printers and internet connections.

A star Topology is used in this experiment, where all end devices are connected to a central network switch. Every device on a LAN must be assigned an IP address within same network range to communicate with each other.

Network diagram:

The LAN was implemented using star Topology as shown below, with 4 PC's connected individually to a central switch

enable
configure terminal
interface vlan1
Hostname LAN-SW
ip address 10.1.1.5 255.0.0.0
no shut down
exit



IP Address configuration Table

Device	IP Address	subnet mask	connected To
PC0	10.1.1.1	255.0.0.0	Switch port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0/2
PC2	10.1.1.3	255.00.0	Switch port Fa0/3
PC3	10.1.1.4	255.00.0	Switch port Fa0/4
Switch	10.1.1.5	255.0.0.0	VLAN 1 interface

Implementation in cisco packet Tracer:

Since physical hardware was limited, the LAN was also implemented and verified using cisco packet Tracer.

Step 1: Place the devices

1. Open Cisco packet Tracer
2. From the device category panel, select End devices
3. Select switches and drag 1 switch

Step 2: Cable the devices.

4. Click connections → Select copper straight-Through cable.
5. Click on each PC, then click switch
6. Repeat for all four PCs. Green link lights confirm a successful physical connection.

Step 3: Assign IP Address to PCs

7. Click a PC → Desktop tab → Ip configuration
8. Enter the static IP Address and subnet mask as per IP for each PC.

Step 4: Configure the switch management IP (Optional)

9. Click the switch → CLI tab
10. Enter the following commands:

```
configure terminal
Interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

Step 5: Verify connectivity using Ping

11. Click a PC → desktop → Command prompt
12. Use the ping command to test connectivity
13. successfull "Reply from" messages confirm connectivity.

Step 6: Demonstrate Data sharing.

14) Add a server device
15. Assign it in IP address → IP configuration tab
16. Open the server's services and enable HTTP service
17. From another PC, open web Browser and enter IP address to load the page.

Step 7: Observe packet Flow in simulation mode

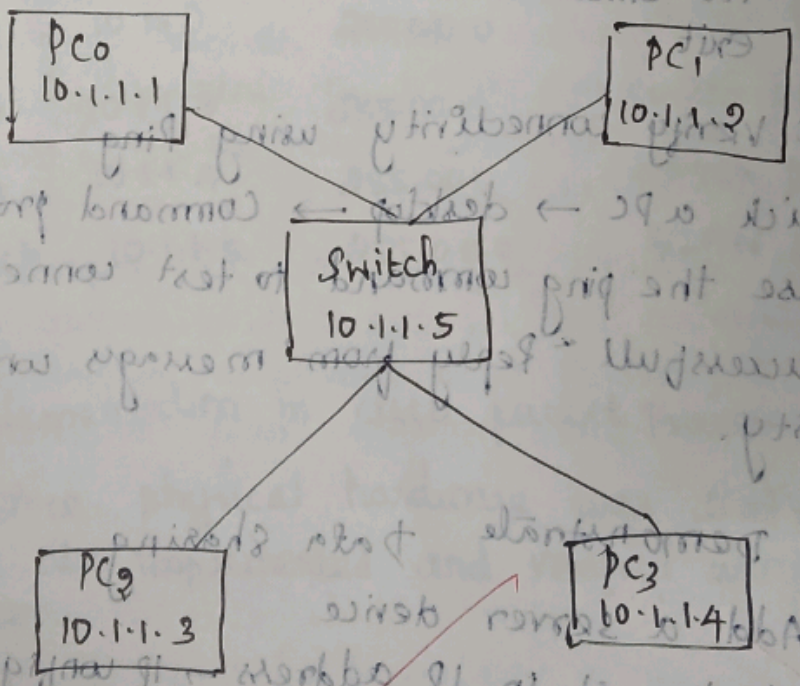
18. Switch to simulation mode

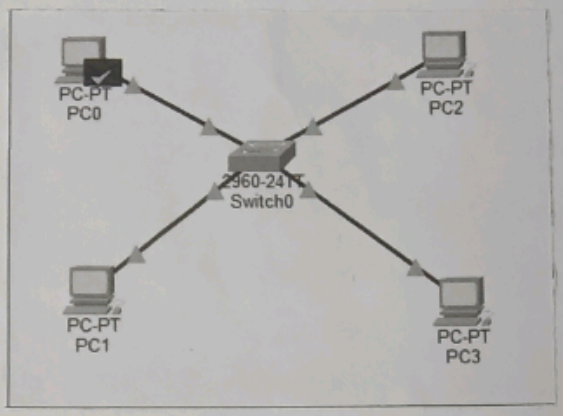
19. Send a ping between two devices and play controls to step through the packet as it travels hop-by-hop, MAC-forwarding

Observation:

All four PCs were successfully connected to the switch in a star topology and assigned unique static IP addresses. within the same subnet (10.0.0.0 \ 255.0.0.0) ping tests between all devices returned successfully.

Network diagram:





Result:

Thus the setting up of Local Area Network (LAN) is implemented and result is obtained